

How to Consider AI Technology Systems as a Part of Human Normative Systems

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Abstract— This paper considers how to understand human normativity and develops an argument against standard philosophical approaches to normativity. It rejects norms as regulations and norms as regularities in favor of a concept of norms that is explained in terms of ongoing existential human concerns. The latter is developed out of recent post-Heideggerian considerations about human practices. This conception of norms is then linked with a systems-theoretical understanding of human social relations, before considering how AI systems will fit into human social systems. It is argued that human technologies are culturally constructive such that their social integration changes who (and how) humans are. The prospective adoption of AI-based systems is assumed, and it is argued that this adoption will increasingly force humans to face up to their ongoing normative concerns so that AI-systems can integrate into human culture. In other words, trustworthiness and responsibility are problematic modern concerns that will keep evolving and showing up in new ways as technologies progressively disrupt human culture.

Keywords— *Normativity, trustworthiness, systems theory.*

I. INTRODUCTION

In the following essay I characterize a conception of normativity that rejects taking norms as either rules governing practices or regularities of behavior found in practices. I explain the view developed by Joseph Rouse such that normativity must be conceived in terms of how we hold one another accountable to what is at issue and at stake in ongoing practices. I introduce the idea of a social systems theory that might be used to analyze norms whilst maintaining Rouse's insights. Together, I believe the insights from Rouse into human normativity and its connection to a social systems theory enables a novel way to consider how AI systems can be integrated into human normativity systems. The second part of this essay is an introduction to considerations about trustworthiness when we conceive of AI and human systems integration in this way.

II. A PHILOSOPHICAL GROUNDING OF NORMS AND NORMATIVITY

'Normativity' is a deeply problematic concept. However, in recent analysis of the philosophical debate about normativity, it is possible to conceive of sides being taken around how theorists ought to view 'norms'. On the one hand, there are normativity sceptics who deny that any satisfactory account of normativity has yet been found. Some of these sceptics prefer to take the view of naturalist social science in accounting for the possibility of everyday reasoning and meaning. They say that we simply have causal explanations of the world, and that empathy plays the role that connects us with the reasoning and meaning of others.

On the other hand, there have formed two main 'normativist' camps of thought that claim normativity is what gives reasons their force and makes words meaningful. These camps also claim that it is normativity that makes sense of why laws and rules are binding. They both attest to there being a non-causal normative domain that explains why how understand and use concepts like right/wrong, true/false, correct/incorrect, valid/invalid.

The first normative camp takes the view that regularities in behavior or practices are what account for normativity. They claim that agents repeat performances in ways that are objectively, or causally describable, such that they exhibit a regularity. Norms, quite simply, are the interpretation of causal regularities. Robert Brandom describes this as the "regularist" approach to explaining the normativity of practices [1]. And Stephen Turner describes it as the "causal" approach to describing the normative unity of a practice [2]. Joseph Rouse suggests it is this view we see conceived in the work of Bourdieu on the idea of 'habitus' [3].

The second normative camp identifies some behavior as action, where both action and intentional practices are conceived as identifiable by their common semantically contentful presuppositions. Norms, in this case, are rules or regulations that govern intentional practices either implicitly or explicitly. Brandom describes this as the "regulist" approach;

whereas Turner takes it as the presuppositional approach. Rouse suggests the work of Kuhn is sometimes read as an example of this approach, where Kuhn's notion of 'paradigms' are read as "presuppositions shared by all participants in a community engaged in normal science" [4]. We can also find this approach in the work of Honneth, or Stahl, in recent critical social theory, where 'immanent critique' is considered possible on the basis of there being implicit 'mutually recognized' presuppositions constitutive of all social institutions [5].

Unfortunately, it is easy to see how both the regulist and regularist approaches to normativity encounter problems of depth in accounting for normativity as criterial for correctness. They are both able to account for normativity in a limited sense, often tied to the idea of community or way of life. So individual utterances or actions can be correct or incorrect within the context of a community or way of life. But there seems no way, on either the regulist or regularist account, of judging the correctness or normative accountability of practices such that norms can be grounded in objectivity rather than mere conventionality. Rouse explains:

Individual performances could be socially incorrect in the sense that they deviate from what others normally do or from communally accepted norms. The *community's* regular behavior or accepted norms would then be criterial for correctness, however, and could not in the same way be understood as open to correction or criticism [6].

There are two main difficulties present in either regulist or regularist accounts of normativity. Rouse identifies these as: (1) inability to account for how social practices are directed toward or accountable to anything other than their own continuation; and (2), inability to explain how and why belonging to a specific community and acceding to its norms could be authoritative to its members [7]. This leads Rouse to consider that "conceptual understanding involves both normative accountability beyond its own actual performances and open-ended capacities for critical reflection" [8]. This means, effectively, that if there is a normative conceptual domain then it cannot be circumscribed in such a way that agents performing in it are unable to reflectively consider the correctness of either social regularities or presupposed norms that underly it.

Rouse then proposes an alternative way of conceiving of normativity that circumvents the limitations of the regulist and regularist conceptions. Specifically, he argues that "we should not think of the social-normative dimension of practices in terms of either behavioral regularities or accepted norms, rules, or conventions" [9]. His proposal is a conjoined social-biological understanding of conceptual normativity:

On this conception, a practice is maintained by interactions among its constitutive performances that express their mutual accountability. On this normative conception of practices, a performance belongs to a practice if it is appropriate to hold it accountable as a correct or incorrect performance of that practice. Such

holding to account is itself integral to the practice and can likewise be done correctly or incorrectly. If done incorrectly, then it would appropriately held accountable in turn. That would require responding to it in ways appropriate to a mistaken holding-accountable and so forth [10].

What this account of normativity presents is a conception of practice or tradition where continuity or coherence does not depend on either an underlying regularity or governing rule. Normativity is to be explained, in Rouse's view, in terms of three crucial aspects which also show why it is not simply reducible to causal regularities [11]. First there is what I will call the *constitutive "diffracting" interconnection of performances*:

...the bounds of a practice are identified by the ways in which its constitutive performances bear on one another, rather than by the regularities of behavior or meaning that they encompass. One performance responds to another, for example, by correcting it, drawing inferences from it, translating it, rewarding or punishing its performer, trying to do the same thing in different circumstances, mimicking it, circumventing its effects, and so on [12].

Secondly, normatively conceived practices imply *performative output concerns*:

...these patterns of interaction {normatively conceived practices} constitute something at issue and at stake in their outcome. MacIntyre (1980) provides a useful illustration of this point: "If I am a Jew, I have to recognize that the tradition of Judaism is partly constituted by a continuous argument over what it means to be a Jew" (p.62). Judaism, like any other significant tradition, exhibits no elements shared throughout its history. What it is to be considered a Jew is instead at issue in the practices of Judaism in all their historical complexity; what is at stake in those practices is the difference it would make to resolve the issue in one way rather than another. But that difference is not already settled, and there is no agreed-upon formulation of what the issues and stakes are. Working out what is at issue in these practices and how the resolution of that issue matters is what the practice is about [13].

Thirdly, normatively conceived practices are *perspectivally variant* or *open-textured*. Rouse explains this feature of practices in terms of Brandom's account of the essentially perspectival objectivity of conceptual norms:

Each perspective is at most *locally* privileged in that it incorporates a structural distinction between objectively correct applications of concepts and applications that are merely subjectively taken to be correct. But none of these perspectives is privileged *in advance* over any

other.... Sorting out who *should* be counted as correct, whose claims and applications of concepts should be treated as authoritative, is a messy retail business.... [T]here is no bird's-eye view above the fray of competing claims from which those that deserve to prevail can be identified.... (Brandom, 2004, pp. 600-1, emphasis added) [14].

Rouse's conception of normativity, in succinct form, is that normativity must be conceived "in terms of how we hold one another accountable to what is at issue and at stake in ongoing practices" [15]. He sees two distinct advantages to conceiving normativity this way, in which we need neither: (1) "posit underlying unity as a condition of understanding" [16]; nor, (2) assume we can identify a "special domain of rules, norms, values, transcendental conditions, natural law, or any other grounds to which philosophers or anyone else has distinctive access" [17].

Together, the *constitutive "diffracting" interconnection of performances*, that are *open-textured* and that each have *performative output concerns*, means that we can describe how normativity arises as the outcome of contextual performances involving critical reflection:

Critical reflection on meaning, knowledge, or social life instead arises from within our practices of communication, understanding, production and exchange, and governance, that is, from the midst of our complex causal intra-actions in partly shared circumstances. Such reflections attempt to express what is already at issue and at stake in those practices, and they are accountable to the very issues and stakes they seek to articulate. There is no god's-eye view that offers a definitive standpoint from which to discern what those stakes really are. There are only ongoing efforts to forge a viable future together from within a shared but contested past and present [18].

I find Rouse's conception of normativity compelling, but it means a question emerges about how we are to conceive of human social ontology. In other words, how are we to explain the social ontology that supports a conjoined social-biological understanding of conceptual normativity. If normativity is explainable partly in terms of biology, there ought to be a theory which suggests how human culture is a specifically biological expression. With such a theory we might then deepen Rouse's account of how we hold one another accountable to what is at issue and at stake in ongoing practices.

One contender is a systems theory that studies complex systems to establish principles of behavior that can be applied to nominal 'units'. We could conceive of such 'units' in the same way as systems theoretical scientists who study ants, bees, fish, or even cells that form tissues [19]. In this way the practice performances referred to by Rouse might be considered both at the normative level mediated by practices of communication, and at a causal systems' level that explains the complexity of

our causal intra-actions, as Rouse says, in partly shared circumstances.

A theory of human causal intra-action that I find particularly attractive as a social-ontological system is explained by Bowen family systems theory. Just as biologists consider complex systems processes of, say, harvester ants, Bowen theory identifies system process in the human family, considered as a basic unit of human biological and cultural form of life. In fact, Leann Howard argues that "the study of the complex systems processes of harvester ants, when compared to processes in the human family, shows primitive indicators of differentiation and provides support for the idea that the same natural forces of individuality and togetherness govern both species" [20]. Howard notes that what connects Bowen systems theory with other biological disciplines that study organic behaviors is identification of functional units that can be described as emotional systems. Regulative similarities are identifiable whether studying harvester ants or the behavior of human beings when each are considered as emotional systems.

As disciplines in the study of organic systems complexity continue to interact, the hypothesis has been developed that biological evolution is not so much simply 'natural' selection but selection in terms of forces operating at many levels, including "genes, chromosome, whole genomes, whole epigenomes, cells, developing tissues, kin groups, societies, communities, and, of course, organisms" [21]. Howard points to several other findings as evidence of acceptance of this view, saying:

A review article in *Science* follows a similar theme by proposing a move from a simple cause-effect model of evolution, natural selection, to a system model based on reciprocity that is the "tracing of causal influences through systems" [22].

Michael Meaney, in a review article for the journal *Child Development*, discusses the interdependence of gene and environment in individual differences in "inherited" stress-activity across generations through maternal care [23]. Maternal effects are being studied in a range of species, from insect societies to the human.

A professor of biology at Stanford University, Deborah M. Gordon, states clearly the complex systems processes of living systems:

Physiological, social, and ecological processes all operate simultaneously and none is more important or fundamental than another. Linking levels of organization is central to any study of social behavior. For humans and social animals, an individual's behavior is always embedded in a social world [24].

[25]

Bowen's family systems theory is generally considered to treat the human family as an organism by considering the reciprocity between members of a family system. The family is considered an emotional unit that is created by interaction processes where the functioning of one member of the family system is established by reciprocal interaction with other family members. I suggest that this family interaction is better understood not as reciprocal interaction, but with Rouse's more nuanced conception: in terms of how we hold one another accountable to what is at issue and at stake in ongoing practices. This would mean that normativity, considered as the *constitutive "diffracting" interconnection of performances* that are *open-textured* and that each have *performative output concerns*, can also be considered as a domain that humans have developed through forces of evolution. The normative domain, on this consideration, functions biologically as a way of mediating evolutionary selection forces.

What becomes interesting when Rouse's conception of the normative domain is coupled with Bowen's conception of the emotional domain in the complexity of family systems, is that we are able to theorize about the emotional content of the normative domain. But it also means we might consider that the causal regularities identified by system theoretical analysis are simply an expression of what is at issue and at stake in ongoing practices of attempts to understand biological systems' complexity. In other words, systems theory and causal explanation is basic to contexts of functional justification, whereas normative theory (as we understand it through Rouse) is basic to contexts of concern for what is at issue and at stake. Rebecca Kukla distinguishes these contextual interests as a distinction between contexts of discovery and justification, where the former is normative and connected to our rational perceptual capacities [26]. Justification can only take place on the basis of there being a context of discovery such that there is something at issue and at stake, sometimes described as a 'problem situation' [27].

Bowen's family systems theory began as an observational project to study individuals with severe symptoms of psychosis. Bowen was able to observe psychotic symptoms over long periods in an institutional setting and he encountered a breakthrough moment when he became aware of how patients changed their behavior when their mothers visited. Psychotic symptoms appeared as a functional form of communication that operated as double binds on family members, particularly in the triad of child, mother and father. Histories of families were developed to find that severe psychotic symptoms commonly appeared in children of parents who were not able to emotionally regulate their relationships through transparently discursive forms of interaction. The double bind was a feature of 'psychotic' families such that children with psychotic symptoms appeared to be maintaining a functioning equilibrium in the family unit to help the parents be more adaptive. In simple terms, emotionally dysfunctional or immature mothers who were in partnerships with similarly emotionally dysfunctional or immature men required a symptomatic child to enable their family unit to maintain functional adaptiveness. Without the 'rescuing' of the parents

by the symptomatic child, the family unit would dissolve; the parents would separate or become increasingly socially dysfunctional.

When Bowen continued his research with less symptomatic families as well as families free of psychotic transference symptoms, he found similar patterns of milder forms of emotional transference. Family members communicate with triangulation where a dyad establishes contexts of concern for what is at issue and at stake by invoking a third family (or extra-family) member for a practice performance [28]. Often this triangulation functions in terms of purported or actual attempts to persecute, rescue, or express victimization to persecution in relation to others. The overall adaptiveness of families became apparent to Bowen in terms of the intensity of their performances in triangulation rather than the fact that families do triangulate. Maladaptive psychological symptoms were also understood as an outcome of the family system as a unit, rather than specifically in the 'head' or 'mind' of an individual.

If Bowen's family systems theory is correct, then we might propose that it can be cleanly connected with Rouse's explanation of normativity. Psychologically maladaptive families would demonstrate chronic practices that consist in *constitutive "diffracting" interconnection of open-textured performances with performative output concerns* that are not as socially adaptive as families that are considered to have less symptoms. Rouse's account of 'how we hold one another accountable to what is at issue and at stake in ongoing practices' could be applied in family studies to the ways specific families hold each other accountable to what is at issue and at stake in ongoing practices. We might then be able to tie these practices to capacity outside the family unit to participate in practices that have deeper social functions, broadly conceived as navigating through cultural life (at school, work, socializing etc.).

It seems plausible that socially maladaptive families will produce offspring who are similarly socially maladaptive unless they can differentiate from their family through development of 'outside' forms of triangulation such that they can develop new concerns about what is at issue and at stake. We might consider that 'normal families' describe those that produce individuals that can take part to a good extent in forms of life outside the scope of mere family's emotional interests.

The trouble that comes with Bowen's family systems theory is how much it broadens Rouse's account of the normative domain past social and explicitly discursively mediated concerns. It appears to indicate that normative domain as we conceive of it philosophically in terms of individuals coping with the objective world is more fundamentally rooted in family coping. If family coping has the level of emotional intensity found by Bowen in his study of psychotic families, then concern for the objective world appears to be atrophied. There appears to be a hierarchy of emotional needs such that humans, who are tied to emotional systems through biological necessity, can only find higher normative concerns once basic normative concerns are met.

III. HOW TO CONSIDER AI TECHNOLOGY SYSTEMS AS A PART OF HUMAN NORMATIVE SYSTEMS

Often, when Artificial Intelligence is discussed in an ethical setting, what stands out to philosophers is the issue of the requirements that ought to be at stake when programming AIs to be trustworthy to humans. Norms are considered as programmable rules or regulations that in principle, if complex enough, ought to be able to make AIs safe for human interaction. The idea is that programmable “autonomous” agents ought to elicit some sort of normative sensitivity or compliance to rules so that they do not negatively affect human well-being. This seems plausible enough. If we think of AIs as complex technologies that continue to adapt their functioning on the basis of massive amounts of data collection, we are interested in giving them a programmed constraint in terms of rules that would prevent them from being ‘untrustworthy’ or causing harm.

‘Norms’ in these discussions are typically considered as rules and described as ‘normative principles’. So, for example, we hear talk of norms as SLEEC rules – rules that are of a social, legal, ethical, empathetic, or cultural nature. The SLEEC schema is considered to be a generalized way of capturing sub-domains that are important to us within the normative domain. But, on the view I have just articulated, if norms are simply SLEEC rules then they are not what we mean by the norms that are the ground of human normativity.

Following Rouse, I have taken a more satisfactory account of normativity, than either ‘regularist’ or ‘regulist’ accounts, to be that normativity must be conceived ‘in terms of how we hold one another accountable to what is at issue and at stake in ongoing practices’. ‘Norms’, on this account, are “what is ‘at issue’ in a practice and what is ‘at stake’ in how it continues” [29]. How then are we to conceive of programming AIs to be trustworthy in terms of our norms, when the very idea of ‘our norms’ is something that is an ongoing concern mediated by normative practices?

I believe there is an interesting way to think about AIs as human technologies such that they are able to be conceived as part of ‘our’ normative practices. Consider the practice of ‘hammering’ specifically using a hammer. If we consider the norms of hammering such that the technology of the hammer follows them, we consider what is at issue in the practice of hammering and what is at stake in how it continues. In general, just as we perhaps observe a child using a hammer, we can define rules for how it must be held, how it works best, and what it is generally used for (eg. hitting nails into wood, shaping metal, breaking up materials, etc.), as well as what it is not to be used for (eg. as a weapon).

With the training of a child, the hope is that they will at some point ‘get’ what is at issue and at stake in the practice of hammering. We suppose they can ‘get it’ because, in Rouse’s terminology they ‘get’ the *constitutive “diffracting” interconnection of performances*. They learn to understand how other human performances respond to performances of hammering, “for example, by correcting it, drawing inferences from it, translating it, rewarding or punishing its performer,

trying to do the same thing in different circumstances, mimicking it, circumventing its effects, and so on” [30].

The question that emerges at this point is whether AIs can be trained like the way we train children in simple functional practices. There seems no reason to doubt that we can, as current AIs seem to excel at functional or instrumental practices and respond to other performances over which they are also collecting massive amounts of data. Indeed, AIs at this level may be so trainable that they may emerge as something like the horrible little bureaucratic ‘know-it-all’ children who are functionally very proficient at some things, but do not understand the nuances of human social interactions. This also seems to be the case currently, if we consider some AI chat tools, as we see some chat AIs express the ideological views of their programmers.

The issue of the relationship between AI intelligence and human normativity must then go a bit deeper than simple cases of AI trustworthiness on functional instrumental tasks. I can imagine an AI assistant at this level will soon be no less trustworthy than bureaucratically trained children or slaves are on most simple tasks. The issue with bureaucratically trained children and slaves is simply that they either do not have, or are not permitted to have, what we call ‘a mind of their own’. They will be able to ‘get it’ with simple functional tasks, where ‘getting it’ means ‘reckoning’ as used by Brian Cantwell Smith in *The Promise of Artificial Intelligence* (2019) [31]. He explains:

I use the term “reckoning” for the representation manipulation and other forms of intentionally and semantically interpretable behavior carried out by systems that are not themselves capable... of understanding what it is that those representations are about – that are not themselves capable of holding the content of their representations to account, that do not authentically engage with the world’s being the way in which their representations represent it as being. Reckoning, that is, is a term for the calculative rationality of which present-day computers (including second-wave AIs) are capable [32].

‘Having a mind of one’s own’, then, would be something beyond ‘reckoning’, which Smith refers to as having the capacity for normative “judgment” [33]. Smith describes ‘judgement’ as the normative ideal to which we should hold full-blooded intelligence – “a form of dispassionate deliberative thought, grounded in ethical commitment and responsible action, appropriate to the situation in which it is deployed” [34]. This is a more common way of expressing what Rouse explains as the normative practice of considering what is at issue in a performative practice and what is at stake in how it continues. Ethical commitment is the commitment of concern for what is at issue and at stake, just as responsible action is when this commitment of concern is maintained and applied in the right circumstances.

The difference between instrumental ‘reckoning’ and normative ‘judgment’ seems to be that whereas both can be

advanced to deal with the *constitutive* “*diffracting*” *interconnection of performances*, and both can be *perspectivally variant* or *open-textured*, what distinguishes proper normative judgment is its performative output concern, its concern with what is at stake in how practices continue and what makes salient issues relevant. Bureaucratic little children are simply kids who have become adept at pretending to be adult, at playing adult, without realization of what really is at stake in the performance of cultural practices. At best they learn to identify issues through mimesis.

The question that now emerges is whether an AI could have a mind of its own and remain trustworthy. Indeed, do other people remain trustworthy when they are thought to have ‘minds of their own’? If we try to imagine AIs being developed so that they have concerns for what is at stake in the performance of practices, there seems to be an issue of how to understand what that means. It makes us realize that it is difficult to understand what it means for humans to have concerns for what is at stake in the performance of practices. But a social ontology such as offered by Bowen systems theory can help. Rouse explained performative output concerns using MacIntyre’s explanation of human concern with traditions. It is not difficult to expand the idea of concern with traditions into the idea of concern for family systems, or kinship systems, that are developed from the systems in which we find ourselves in our family-of-origin.

When we ask again whether people are trustworthy when they are thought to have minds of their own, we can anticipate replies. Indeed, we can anticipate a broadly Aristotelian reply: we can trust people if they indicate they have been brought up well such that they are virtuous. This means, I think, that we trust people who show functional reckoning over simple functional matters, but we form a deeper sense of trust with people who indicate knowing what is at issue and at stake. It seems unproblematic to think that we might trust AIs in terms of their reckoning abilities, but we would be foolish to think that they share our human concerns if they have not been brought up as humans, in family systems that can triangulate into wider forms of social organization.

This means, as Smith says, that we might well be glad that AI systems can function in situations of moral gravity: “When landing in San Francisco, I am glad the airplane is guided by sophisticated computer systems, rather than by pilots peering out through the fog looking for the airport” [35]. What is important, as Smith also believes, is that we learn to how to live with technologies that can interact with us and to large extents take over our functional requirements in complex practices. But we must not lose perspective to see that, at best, they are a technological supplement for our ongoing human concerns.

What is at issue, perhaps, considering the rapid development of AI systems with deep-learning capabilities, is that we do not lose sense of what we understand as human common sense. ‘Common sense’ seems a simple and well-known term for expressing normative awareness, which is to say, cultural familiarity in considering what is at issue in performative practices and what is at stake in how they continue. Thus, we can say that foreigners seem to lack

common sense in the simple sense of them not knowing what is at issue and at stake in local situations. Consider the phrase: “Only mad dogs and Englishmen go out in the midday sun”. Presumably it is not meant such that Englishmen are mad, but only that they’ve been known to lack common sense in territories with hot climates. But another use of common sense is precisely to indicate madness, or what we have been calling socially maladaptive behaviors and thinking.

Let us now consider the situation of a world where Rouse’s theory of normativity is correct, Bowen systems theory is correct, and Smith’s theory of AI reckoning capabilities is correct, such that AIs are functionally advanced assistants but fall short of joining humanity by sharing our concerns. We might say that AIs cannot join the normative space of reasons that humans maintain through ongoing practice-mediated judgment. But note that we have developed an interesting picture about humanity as a culture developed out of triangulation from family systems. In some cases, and perhaps many cases across different contexts, humans are maladaptive. This means that although we can occupy a normative space of reasons, in many circumstances and to greater or lesser degrees we fail to show common sense in localized contexts.

Common sense is a skill developed through familiarity with the performance of practices over which we are existentially concerned with what is at issue and what is at stake. The dangers we face are often due to lacking common sense, lacking judgment, lacking familiarity with what it means to be existentially concerned. How are we to explain Dunning-Kruger cognitive bias, where individuals with low competence in a specific area tend to overestimate their abilities and knowledge, while those with high competence may underestimate their skills? It is said that this bias arises because individuals lacking knowledge are often unaware of their own lack of expertise, leading them to overrate their abilities.

But we might add to this explanation that generally many of us misunderstand the kind of ability it takes to be human, normative ability. We may assume to a great extent that ability is functional and that we can understand because we have a sense of how things seem to be governed by regulations and rules. But in doing this are we really that different from the bureaucratic child? The real skill in being human is in knowing what is at issue and at stake in practices and how we hold one another accountable to what is at issue and at stake in ongoing practices.

The danger of using advanced AIs then, may be similar to the danger of using advanced information services such as contemporary news platforms. We seem to rely on them as if we are fully capable and blessed with ubiquitous common sense. We don’t seem to appreciate that we have a very difficult time understanding our own abilities for concernfulness that are rooted in the way of life we have developed by triangulating within a family, and if successful, differentiating some way out of it into society too. And we are also terrible at holding one another accountable.

In other words, the issue is not the trustworthiness of AIs; they have the potential to be incredibly trustworthy in functional terms. The real issue is that we, humans, are

woefully untrustworthy because we do not yet understand the complexities of our normativity such that we can understand how to assess and develop our common sense. We seem to be able to assess common sense in some practices, perhaps those with which we have been familiar as a matter of tradition and deep concern. Yet we also seem to lack common sense to enormous degrees when engaging in practices that are unfamiliar; new practices based on new technologies tend to fall in this bracket. The Dunning-Kruger effect shows that we are happy to ‘fly blind’ with false confidence.

An added issue comes with the training of AIs, or the setting of parameters so that AIs can self-train. The standard for intelligence might well be explained as demonstration of common sense. We have yet to see AIs demonstrate common sense, precisely because they cannot share human concerns. But what we are tending to neglect, in part because of our Dunning-Kruger cognitive bias, is that ‘common sense’ is a highly developed set of human skills and only small fractions of common sense are shared across human collectives. It takes incredible commitment to training, involving learning to work together in the performance of shared practices, for humans to acquire shared common sense.

We assume others share our common sense in simple ways, sometimes by seeing that they speak our verbal language, but this is often to assume far too much. A shared language can hide the fact that humans use that language to mediate incredibly different sorts of practices oriented by very different concerns for what is at issue and what is at stake. This is effectively what is diagnosed in socially maladaptive families by Bowen systems theory; psychotic families are seen to use standard verbal signifiers to signify in complex, emotionally laden, often double-binding ways, that are absolutely perplexing to onlookers.

There seems a social effect that might also be tied to our Dunning-Kruger biases such that we cannot seem to reflect on how *different* human cultures in fact are, such that we cannot believe others don’t interpret the “facts” of the world in the same way we do. What then happens when individuals in these diverse cultures use advanced AI technologies to adapt how they function in shared environments without grounding in shared concerns?

One idea is that the differentiation we see between the implied emotional schemas of family systems might become more prominent. Consider again emotionally dysfunctional families studied by Bowen, where psychotic transference conditions determine some individuals to use psychotic symptoms for intra-family connection. What happens when complex technology is introduced into systems with that level of social functioning? It seems plausible that systems could adopt the tech without adapting their social function. This may mean AI will prompt new types of expression of emotional illness. Some argue we already see this effect with the introduction of screen technology and digital communication.

This does not mean that trustworthiness is a problem of AIs. It is not even that the issue is of the trustworthiness of family systems, or other kinship systems, that emotionally regulate in ways that are antisocial for the majority of others. The issue of

trustworthiness must be directed at the poverty of our philosophical understanding such that we fail to understand the complexity of human normativity. In some sense, the introduction of advanced AI technologies is like giving power tools to people who do not really know how to use them. Yet, haven’t we always done this? Perhaps, yes, but we have not tended to introduce new technologies that are truly game-changing across so many human practices at a global scale until relatively recently. And we seem particularly inept in reflective appreciation that this is a problem that can produce consequences we do not know how to handle.

There is a fairly simple schema to keep in mind what happens to social organization with the introduction and subsequent socially invested adoption of new technologies. Consider that technology adoption alters how we self- and socially-organize. Heavily disruptive technologies alter how we structure our time, for example, how we structure what we do throughout a given day. As technologies interact our social organization becomes increasingly complex. We alter our world to such an extent that it alters us; we behave differently. The issue is that our changed behavior has an effect on our social and emotional regulation, which leads us to introduce technologies in different ways just to cope. In some cases, this dialectic of technology and social organization leads to increased inability to cope in certain areas of life, which leads to many of the negative outcomes that litter humans’ history: emotional upset, social disintegration, and violence.

Perhaps oddly, as novel and game-changing as AI technologies promise to be (and already are), our issues around AI trustworthiness are issues in knowing ourselves better. We have been ‘here’ as a species many times before. Game-changing technology does not merely disrupt markets and restructure human being in terms of our time is structured, it forces us to face ourselves, and to face our normative commitments. Rouse describes normativity in terms of our ongoing need for concern with what is at issue in our practices and what is at stake in continuing them.

The critical social theorist’s challenge around trustworthiness and AI, is to enable means for theoretical reflection on how integration of AIs with our practices changes what is at issue, what is at stake in their continuance, and thus how we ought to focus our concerns. It is the rapid market disruption from AI introduction that is perhaps what will shake our sense of technological trust. When our social organization changes so fast, historically we have shown that it is very difficult to refocus our social concerns such that we can keep up. In a strong sense, with the increasing rapidity of social change due to pervasive introduction of socially disruptive technologies, we must all learn how to become critical social theorists – we must learn to take ourselves philosophically seriously.

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- [5] Honneth, A. (2014) *Freedom's Right – The Social Foundations of Democratic Life*, Translated by Gandel, J. Cambridge: Polity Press; Stahl, T. *Immanent Critique*. Translated with John Baptiste Oduor. London: Rowman and Littlefield.
- [6] Rouse, J. (2015) *Articulating the World Conceptual Understanding and the Scientific Image*. Chicago: University of Chicago Press, p.162; my emphasis.
- [7] *Ibid.*
- [8] *Ibid.*
- [9] *Ibid.* p.163
- [10] Rouse, J. (2006) 'Social Practices and Normativity', in *Philosophy of the Social Sciences* Volume 37 Number 1 March 2007 46-56, Sage Publications, p.48
- [11] *Ibid.* p.49
- [12] *Ibid.* – Rouse explains the reflexivity of performances in terms of 'diffraction', commenting: "Donna Haraway (1997) and Karen Barad (2007) offer a suggestive metaphor for these relationships among performances of a practice by talking about them as "defracted" through one another, instead of invoking the more familiar metaphor of "reflexivity"."
- [13] *Ibid.* p.50
- [14] *Ibid.* pp.50-51
- [15] *Ibid.* p.54
- [16] *Ibid.*
- [17] *Ibid.* p.55
- [18] *Ibid.* p.54
- [19] (Howard, 2014; Couzin, 2007; Greene and Gordon, 2007; Seeley, 2010)
- [20] (Howard, 2014, p.99) - Note that conceptions of regularities and governance by rules are invoked in systems-theoretic language, but they do not refer to normative regularities or governance, but simply causal regularities and principles expressive of rules characterizing these causal regularities.
- [21] (Fodor and Piattelli-Palmarini, 2011, p.57)
- [22] (Laland et al., 2011, p.1516)
- [23] (Meaney, 2010)
- [24] Gordon, (1999, p.96)
- [25] (Howard, 2014, p.100)
- [26] Kukla, R. (2006) 'Objectivity and Perspective in Empirical Knowledge' in *Episteme: A Journal of Social Epistemology*, Volume 3, Issue 1-2, 2006, p.84
- [27] Thomas, A. *Value and Context*. Oxford: Oxford University Press, p.284
- [28] Normative 'practice performances' appear to be the same as the 'games' identified by Eric Berne in transactional analysis. See: Berne, E. (1964) *Games People Play – The Psychology of Human Relationships*, Grove Press.
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